



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav of Information Technology
Department of Computer Science and Engineering

Industrial Training Report



Industrial Training Report Submitted
towards partial fulfillment of the degree of
Bachelor of Technology

Year 2025-26

ENROLL – 22100BTCSE11628

Guided By

Prof. Priyanka Gupta

Submitted By

Muskan Keer



Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore
Shri Vaishnav of Information Technology
Department of Computer Science and Engineering

Shri Vaishnav Institute of Information Technology



Certificate

This is to certify that Mr. **MUSKAN KEER** has satisfactorily completed the Industrial Training towards the partial fulfillment of the degree in Bachelor of Technology (Department) awarded by Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore for the academic year 2025-26.

Internal Examiner

Head of the Department

Dr. Anand Rajawat
Director, SVIIT



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Abstract

The 45-day Industrial Training at **Universal Informatics, Indore**, strengthened my foundational skills in programming and problem-solving. The training focused on **C, C++, Data Structures & Algorithms, SQL, and MySQL**, using industry-standard tools like **VS Code** and **MySQL Workbench**. Through hands-on coding practice, LeetCode exercises, classroom sessions, and regular assessments, I developed strong technical competence essential for placements. This report summarizes the training schedule, technical work, learning outcomes, and professional skills gained during the program.

Acknowledgement

I express my sincere gratitude to **Universal Informatics, Indore** for providing me the opportunity to undergo 45 days of Industrial Training starting from **3rd July 2025**. I am deeply thankful to my industrial mentor **Mr. Jitendra Malviya** for his continuous guidance, support, and valuable insights throughout the training period.

I would also like to thank the entire training team for providing structured sessions on **C, C++, SQL, MySQL, and Data Structures**, and for helping me strengthen my coding and problem-solving skills for placement preparation.

Finally, I extend my heartfelt thanks to **Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore**, faculty members, and my department for offering such an important industrial exposure as part of the curriculum.



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CHAPTER NO. 1

INTRODUCTION

Industrial Training is an essential component of the B.Tech (Computer Science & Engineering) curriculum, aimed at enhancing the practical knowledge, technical competency, and professional skills of students. As a final-year CSE student at *Universal Informatics College, SVVV Indore*, this 45-day industrial training provided me hands-on exposure to real-world software development practices.

The training started on **3rd July 2025** and covered essential areas such as **C Programming, C++ with OOPs, Data Structures & Algorithms (DSA), SQL, MySQL**, and problem-solving using **LeetCode**. The training was conducted using tools such as **VS Code, GCC Compiler, and MySQL Workbench**.

1.1 OBJECTIVES OF THE INDUSTRIAL TRAINING PROGRAM

The primary objectives of this industrial training were:

- To bridge the gap between academic learning and industrial practical requirements.
- To provide hands-on experience in programming languages such as C and C++.
- To strengthen understanding of Data Structures and Algorithms through practical problem-solving.
- To build familiarity with database management systems and SQL queries.
- To enhance coding efficiency, debugging skills, and logical thinking required for placements.
- To help students apply theoretical knowledge to practical software development tasks.
- To improve industry-oriented skills such as teamwork, communication, discipline, and professionalism.

1.2 SCOPE OF THE INDUSTRIAL TRAINING PROGRAM

The scope of this training program included:

- Learning and practicing programming fundamentals using C.
- Understanding Object-Oriented concepts through C++ and applying them in real programs.
- Solving DSA problems on arrays, strings, linked lists, stacks, queues, trees, and dynamic programming.
- Writing optimized SQL queries for data retrieval, insertion, update, and deletion.
- Working with MySQL databases using MySQL Workbench.
- Developing industry-required logic-building and analytical skills for placements.
- Preparing students for technical interviews through daily coding questions and practice tasks.



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1.3 DURATION OF TRAINING

The total duration of the industrial training was:

- **45 Days**
- **Start Date:** 3rd July 2025
- **End Date:** (Automatically implied after 45 days)
- Conducted in offline mode with regular sessions, assignments, and practice tasks.

1.4 IMPORTANCE OF INDUSTRIAL TRAINING

Industrial Training plays a crucial role in shaping students into industry-ready professionals. It helps in:

- Gaining practical exposure to programming and software development tools.
- Understanding how theoretical concepts are applied in real scenarios.
- Improving technical, analytical, and problem-solving skills.
- Learning to work in structured and disciplined environments.
- Preparing for campus placements by practicing real interview questions.
- Enhancing confidence and communication skills.

1.5 BENEFITS DERIVED FROM THE TRAINING

During this training, I gained several academic and professional benefits:

- Strong foundation in **C/C++ syntax, logic, and problem-solving**.
- Improved understanding of **OOP concepts** such as classes, objects, inheritance, polymorphism, and templates.
- Ability to solve **Leetcode DSA problems**, enhancing coding confidence.
- Hands-on experience in running SQL queries and managing databases.
- Better clarity of data structures used in interviews.
- Experience using **VS Code** and **MySQL Workbench**, similar to industry tools.
- Enhanced time-management, discipline, and professional behavior.



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1.6 OBJECTIVES OF THIS REPORT

This report aims to:

- Document the complete learning experience gained during the 45-day industrial training.
- Present details of the topics learned, assignments completed, and tools used.
- Highlight the practical exposure and technical skills developed.
- Provide a structured explanation of the project tasks, sample code, and screenshots.
- Demonstrate the learning outcomes and readiness for future professional roles.



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CHAPTER NO. 2

TRAINING DESCRIPTION

The Industrial Training program provided a structured and well-planned learning environment aimed at strengthening the core technical skills required in the field of Computer Science & Engineering. The training was conducted, who ensured that every student understood the concepts thoroughly through practical implementation.

This chapter presents a detailed explanation of the training environment, teaching methodology, tools used, and the skills developed during the 45-day program. The training covered **C Programming, C++ with OOPs, SQL & MySQL**, and **DSA with LeetCode problem-solving**, which are essential for both academics and campus placements

2.1 OVERVIEW OF TRAINING ENVIRONMENT

The training was conducted in a professional classroom-lab environment equipped with systems, projectors, and internet access. Each student worked individually on their workstation using **VS Code** and **MySQL Workbench**. The mentor ensured that every concept was explained with examples and followed by hands-on practice. Doubt-solving, regular assignments, and coding tests created a disciplined and productive environment. This setup helped students learn in a real industry-like atmosphere where coding, testing, debugging, and optimization were practiced daily.

2.2 C PROGRAMMING TRAINING

The training started with the fundamentals of **C programming**, the building block of all programming languages. Students learned about variables, data types, operators, conditional statements, loops, arrays, strings, pointers, structures, and file handling. Special focus was given to logic-building because C helps improve problem-solving skills. Simple to advanced programs were practices such as calculator programs, pattern printing, array operations, and string handling. This module created a strong programming foundation and prepared students for advanced concepts in C++ and DSA.



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2.3 C++ & OOPs TRAINING

After mastering C basics, the training shifted to **C++** and **Object-Oriented Programming**. Students learned about classes, objects, constructors, destructors, inheritance, polymorphism, abstraction, encapsulation, templates, and STL (Standard Template Library). Real-world examples were used to teach OOP concepts, making learning easier. Programs were created using vectors, maps, stacks, queues, and file handling in C++. C++ training also improved students' understanding of how modern applications are built using OOP principles.

2.4 SQL & MYSQL TRAINING

A major part of the training focused on database management using **SQL** and **MySQL Workbench**. Students learned how to create databases, tables, and relationships. Basic to advanced SQL queries were practiced, including:

- SELECT, INSERT, UPDATE, DELETE
- WHERE, AND, OR conditions
- ORDER BY, GROUP BY
- JOINS (INNER, LEFT, RIGHT)
- Constraints and Keys

Students also created small database projects with proper queries for data storage and retrieval. This training helped build strong command over backend data handling, which is essential for all software development roles.

2.5 DSA & LEETCODE PROBLEM SOLVING

The most important part of the training was **DSA (Data Structures & Algorithms)**. Students practiced problems from **LeetCode** to enhance placement-oriented coding skills. Topics included arrays, strings, linked lists, stacks, queues, recursion, searching, sorting, and basic tree problems. Each problem was explained with logic, dry-run, time complexity, and multiple solutions. This module improved analytical thinking, optimized coding ability, and prepared students for technical interviews at top IT companies.



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CHAPTER NO. 3 – INDUSTRIAL TRAINING

This chapter provides a detailed description of the technical training conducted during the 45-day industrial program. It includes the objectives, technical analysis, tools used, techniques studied, UML diagrams, and sample code that demonstrates the learning outcomes. The training modules—C, C++, SQL, and DSA—were taught with practical implementation to prepare students for real-world software development and placement requirements.

3.1 OBJECTIVES

The main objectives of the industrial training were:

1. To develop strong fundamentals in programming using **C and C++**.
2. To understand and apply **Object-Oriented Programming (OOPs)** concepts.
3. To learn **SQL database management** and perform industry-level queries.
4. To enhance **problem-solving skills** using Data Structures and Algorithms.
5. To prepare students for **technical interviews and placement drives**.
6. To expose students to **real coding environments** using VS Code and MySQL.
7. To improve logical thinking, debugging skills, and coding efficiency.

3.2 ANALYSIS

During the training, the technical structure and learning flow were designed in a progressive manner.

- The first phase focused on **logic building using C**.
- The second phase introduced **C++ with OOPs**, improving design thinking.
- The third phase taught **database handling** with SQL queries.
- The final phase focused on **DSA and LeetCode problems** to build placement-ready coding skills.

This structured approach helped students understand how programming, databases, and algorithms work together in software development. Regular assessments, coding assignments, and interactive discussions made the learning process practical and efficient.



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3.3 TECHNIQUES STUDIED IN DIFFERENT PARAMETERS

1. Programming Techniques

- Modular programming
- Use of loops and conditions
- Functions and recursion
- Pointer-based memory manipulation
- Object-oriented techniques (inheritance, polymorphism, abstraction)

2. Database Techniques

- CRUD operations
- Joining multiple tables
- Grouping and filtering data
- Using keys and constraints
- Query optimization basics

3. DSA Techniques

- Time & Space complexity analysis
- Searching and sorting methods
- Linear & Non-linear data structures
- Stack & queue operations
- Hashing concepts\
- Problem-solving strategies (two-pointer, sliding window, recursion)

3.4 ARCHITECTURE & UML DIAGRAMS

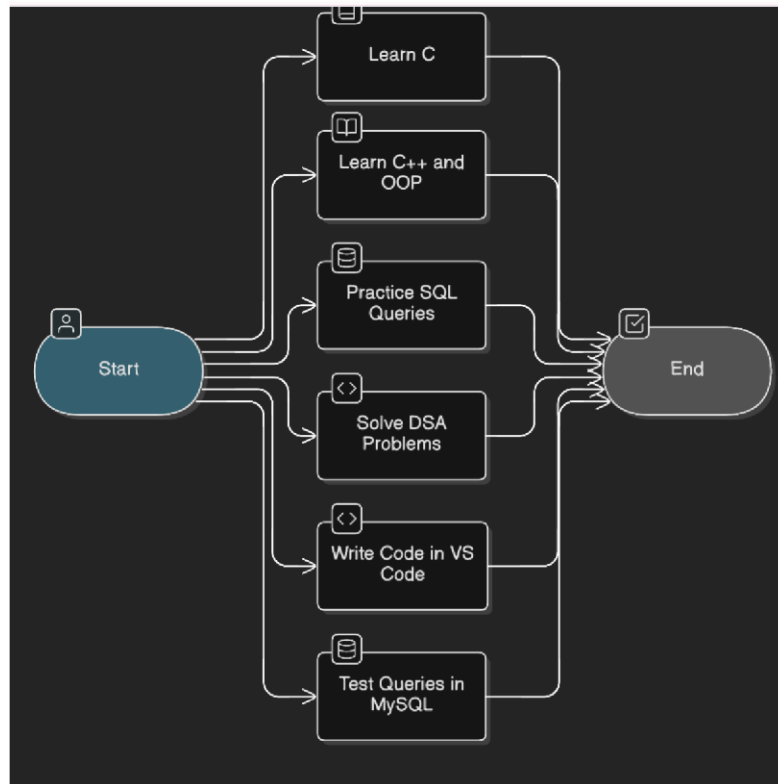
Below is a simple representation of learning flow using UML diagrams.

A. Use Case Diagram for Training Modules

- Student
 - Learn C
 - Learn C++ & OOP
 - Practice SQL queries ○
Solve DSA problems
 - Write code in VS Code
 - Test queries in MySQL



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B. Activity Diagram

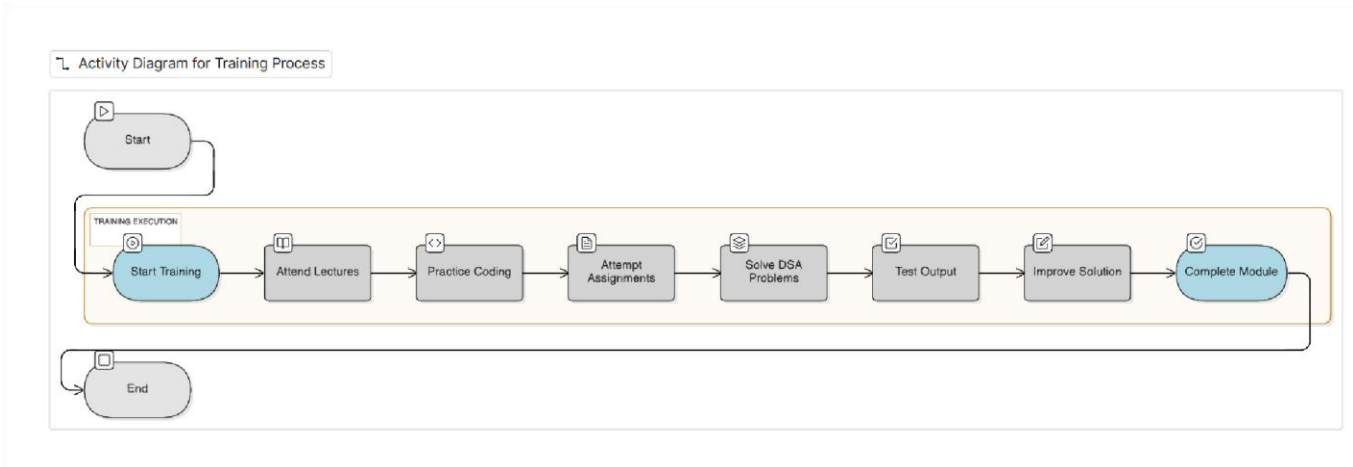
1. Start Training
2. Attend lectures
3. Practice coding
4. Attempt assignments
5. Solve DSA problems
6. Test output
7. Improve solution
8. Complete module
9. End Training



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3.5 DESIGN & SAMPLE CODE

Below are some sample codes that were practiced during training.

Sample Code 1 – C Program (Simple Calculator)

```
#include <stdio.h>
int main() { int
a, b; char op;

printf("Enter first number: ");
scanf("%d", &a); printf("Enter
second number: "); scanf("%d",
&b); printf("Enter operator (+,-
,*,/): "); scanf(" %c", &op);

switch(op) { case '+': printf("Result =
%d", a + b); break; case '-': printf("Result =
%d", a - b); break; case '*': printf("Result =
%d", a * b); break; case '/': printf("Result =
%d", a / b); break; default: printf("Invalid
operator");
}
return 0;
}
```

Sample Code 2 – C++ OOP Program (Class & Object)

```
#include <iostream>
using namespace std;
```



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```
class Student {
public:
    string name;
    int age;

    void display() {        cout << "Name:
" << name << endl;        cout << "Age: "
<< age << endl;
    }
};

int main() {    Student s1;
s1.name = "Nandkishor";
s1.age = 21;

    s1.display();
return 0;
}
```

Sample Code 3 – SQL Basic Commands

```
CREATE DATABASE college;
USE college; CREATE
TABLE students( id INT
PRIMARY KEY, name
VARCHAR(50),
age INT, dept
VARCHAR(20)
);

INSERT INTO students VALUES (1, 'Nandkishor', 21, 'CSE');

SELECT * FROM students;
```



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3.6 SOFTWARE AND TOOLS USED

During the training, the following software and tools were used:

- **Visual Studio Code** (for C/C++ coding)
- **GCC Compiler**
- **MySQL Workbench** (for SQL operations)
- **LeetCode Platform** (for DSA practice)
- **Draw.io** (for UML diagrams)
- **GitHub** (optional practice for version control)

3.7 HIGHLIGHTS OF TRAINING EXPOSURE

- Gained hands-on experience in **C, C++, SQL, and DSA**.
- Understood real-world coding practices and debugging.
- Solved multiple LeetCode problems for placement preparation.
- I learned how to design structured and optimized code.
- Improved teamwork, communication, and analytical skills.
- Got exposure to real industry tools like VS Code and MySQL.



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Chapter No. 4 – Screenshots & Work Done

This chapter contains the practical work completed during the industrial training program. It includes screenshots of coding practice, SQL execution, DSA problem-solving, and outputs generated during the training. All screenshots serve as evidence of hands-on learning and demonstrate the application of concepts studied throughout the modules.

4.1 C Programming Practice

Screenshots in this section show:

- Execution of basic C programs
- Use of loops, functions, arrays
- Debugging steps in the terminal
- Compiling and running code using GCC / VS Code

4.2 C++ & OOPs Programming Practice

Screenshots include:

- Implementation of OOP concepts such as classes, objects, inheritance, polymorphism
- Usage of constructors, destructors, and function overloading
- Execution outputs in terminal or VS Code



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```
1  #include<iostream>
2  #include<cmath>
3  #include<vector>
4  using namespace std;
5
6
7  int main(){
8
9      vector<float> v;
10     float n,m;
11     int ch;
12     cout<<"enter the first numbers:"<<endl;
13     cin>>n;
14     float result =n;
15
16     do{
17
18         cout<<"menu list of operations:"<<endl;
19         cout<<"1. Addition"<<endl;
20         cout<<"2. Substraction"<<endl;
21         cout<<"3. Division"<<endl;
22         cout<<"4. Multiplication"<<endl;
23         cout<<"0. Exit"<<endl;
```

4.3 SQL & MySQL Work Done

This section contains screenshots showing:

- Database creation and table creation queries
- CRUD operations (SELECT, INSERT, UPDATE, DELETE)
- JOIN operations, aggregate functions
- Real-time execution in MySQL Workbench / XAMPP



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MySQL Workbench

Admin-coder x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

coder

Tables

- clg_student
- company
- employee
- employees
- students
- students_copy
- syllabus
- teammember

Views

Stored Procedures

Functions

Query 1 x SQL File 3* teammember clg_student company

1 use coder

2

3 select * from students

4 where s_name not like 'n%'

5

6 select * from students

7 where s_name like '%s%'

8

9 select * from students

10 where s_name like '__n%'

11

12 select * from students

13 where s_name like '__%'

14

Information

Table: company

Columns:

- id int PK
- name varchar(90)
- city char(50)
- address varchar(50)

Output

Action Output

#	Time	Action	Message
1	09:34:20	CREATE TABLE Employees (employee_id INT AUTO_INCREMENT PRIM...	Error Code: 1046. No database selected Select
2	09:34:33	SELECT * FROM coder.clg_student LIMIT 0, 1000	0 row(s) returned
3	09:34:48	SELECT * FROM coder.company LIMIT 0, 1000	6 row(s) returned

MySQL Workbench

Admin-coder x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

coder

Tables

- clg_student
- company
- employee
- employees
- students
- students_copy
- syllabus
- teammember

Views

Stored Procedures

Functions

Query 1 x SQL File 3* teammember clg_student company

15 select * from students

16 where s_name like '__%'

17

18

19 create table students_copy as

20 select * from students_copystudents where 1=2;

21

22 create table clg_student like students

23

24 desc students_copy

25 desc clg_student

26

27

28 DELIMITER \$\$

Information

Table: company

Columns:

- id int PK
- name varchar(90)
- city char(50)
- address varchar(50)

Output

Action Output

#	Time	Action	Message
1	09:34:20	CREATE TABLE Employees (employee_id INT AUTO_INCREMENT PRIM...	Error Code: 1046. No database selected Select the default DB to be used by
2	09:34:33	SELECT * FROM coder.clg_student LIMIT 0, 1000	0 row(s) returned
3	09:34:48	SELECT * FROM coder.company LIMIT 0, 1000	6 row(s) returned

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4.4 DSA & LeetCode Problem Solving

Screenshots include:

- Solved LeetCode problems
 - Code editor view and output
 - DSA concepts such as arrays, strings, linked lists, recursion
-
- Execution results

The screenshot shows a LeetCode problem page for '27. Remove Element'. The problem description states: 'Given an integer array `nums` and an integer `val`, remove all occurrences of `val` in `nums` **in-place**. The order of the elements may be changed. Then return the number of elements in `nums` which are not equal to `val`.' The difficulty is 'Easy'. The solution is marked as 'Solved'. The code editor shows a C++ implementation of the `removeElement` function using a two-pointer technique. The test results show 'Accepted' with a runtime of 0 ms. The input is `nums = [3,2,2,3]` and the output is `val =`.

```
2 public:
3
4
5     int removeElement(vector<int>& nums, int val) {
6         int j=0;
7         for(int i=0;i<nums.size();i++){
8             if(nums[i]!=val){
9                 nums[j] = nums[i];
10                j++;
11            }
12        }
13        return j;
14    }
15};
```

```
int[] nums = [...]; // Input array
int val = ...; // Value to remove
int[] expectedNums = [...]; // The expected answer with correct length.
// It is sorted with no values equaling val.

int k = removeElement(nums, val); // Calls your implementation

assert k == expectedNums.length;
sort(nums, 0, k); // Sort the first k elements of nums
for (int i = 0; i < actualLength; i++) {
    assert nums[i] == expectedNums[i];
}
```

4.5 Tools & Platforms Used

Screenshots showing usage of:



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- Visual Studio Code
- MySQL Workbench / Command Line
- Compiler outputs
- Online platforms like LeetCode

4.6 Summary of Work Done

This section briefly summarizes the hands-on tasks completed:

- More than C programs were written
- C++ OOP-based programs were implemented
- SQL queries practiced
- Multiple LeetCode problems solved
- Code was tested and executed successfully



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Chapter No. 5 – Conclusion

The industrial training program proved to be a highly valuable learning experience that strengthened both theoretical knowledge and practical skills. Throughout the training period, I worked on core programming modules such as C, C++, SQL, and Data Structures & Algorithms (DSA), which improved my logical thinking, coding efficiency, and problem-solving ability. The hands-on practice with real-time coding, query execution, and debugging helped me gain confidence in writing optimized and error-free programs.

The exposure to industry-oriented problem-solving, structured coding practices, and the use of tools like Visual Studio Code and MySQL Workbench enabled me to understand how actual software development environments operate. Additionally, solving LeetCode problems improved my analytical skills and prepared me for technical interviews and placement examinations.

Overall, this training enhanced my technical foundation, improved my work discipline, and helped me understand the expectations of the IT industry. The experience gained through this training will significantly contribute to my future academic progress and professional career, especially in software development and coding-based roles.

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